OXFORD
INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL GCSE
BIOLOGY
9201/2

Paper 2

Mark Scheme

November 2022

Version: 1.0 Final

22BY9201/2/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Assessment Writer.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from aqa.org.uk

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

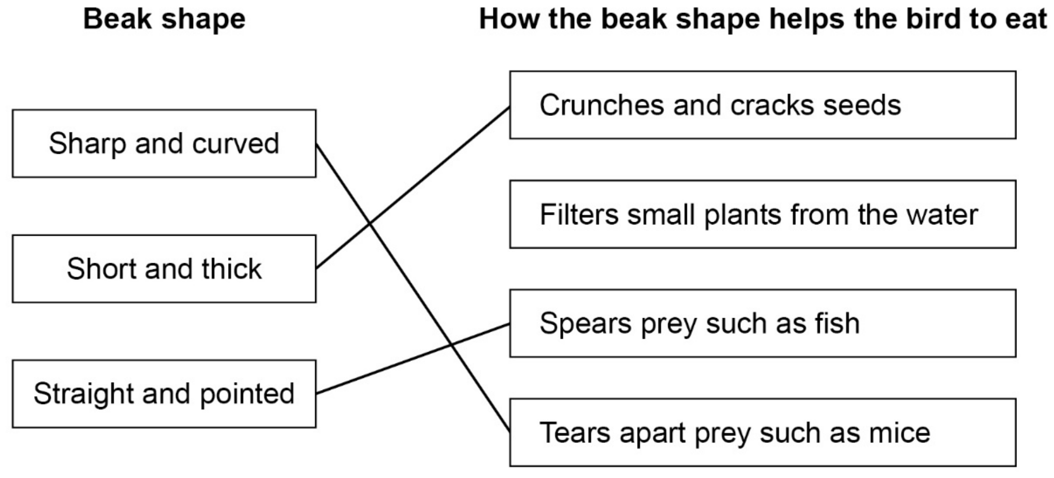
Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	territory		1	AO1 3.3.2c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	do **not** accept more than one line from a box on the left. 		3	AO2 3.3.2f

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.3	Step 1: **B** → Step 2: **D** → Step 3: **A** all correct = **2** marks any 2 correct = **1** mark		2	AO4 3.3.2 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	stopwatch	allow stop-clock / timer ignore paper / pencil / pen ignore mobile phone unqualified	1	AO4 3.3.2 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	bar chart		1	AO4 3.3.2 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.6	any **one** from: • reduced competition for food • less likely to starve	allow provides a variety of nutrients or minerals or vitamins	1	AO3 3.3.2c

Total			**9**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	any **one** from: • root • stem	allow flower or fruit	1	AO1 3.1.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	carbon dioxide		1	AO2 3.2.2ac

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	guard (cells)		1	AO1 3.2.2e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	**B**		1	AO2 3.1.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	**D**		1	AO2 3.1.4a 3.2.2b

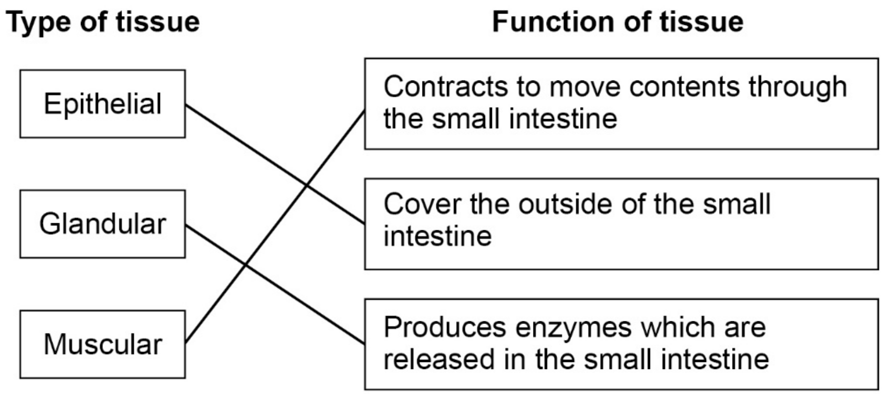
Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	increases the surface area of cells in contact with the air		1	AO2 3.1.4a 3.2.2b 3.1.5bij
	(so gases can) diffuse (into / out of the cells)		1	AO1
	faster / more quickly		1	AO1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.7	$\frac{7}{40}$		1	AO2 3.2.2b 6.3(3,10)
	(=) 0.175 (cm)		1	
	1.75 mm	allow correct unit conversion from incorrect calculation	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.8	(Leaf A / in centre of woodland)			
	receives less light		1	AO3 3.2.1bc
	(so) has a larger surface / area to absorb as much light as possible	allow has more chloroplasts to absorb as much light as possible	1	AO3
	for photosynthesis		1	AO2

Total			**14**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	3 correct = **2** marks 2 or 1 correct = **1** mark do **not** accept more than one line from a box on the left.		2	AO2 3.1.3ab



Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	any **one** from: • wear eye protection / goggles • wear (protective) gloves • wash hands • cover any cuts (on exposed hands and arms) • do not taste / eat		1	AO4 3.2.4 RP3 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	all (wells / iodine) turn blue-black	allow starch present in all wells **or** starch is not broken down	1	AO4 3.2.4 RP3 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	$\frac{10}{1000} \times 2$		1	AO2 3.2.4 RP3 (3,16)
	0.02 (g)		1	
	$\frac{0.02}{60}$		1	
	0.00033 ‘ / 0.00033 … (g/s)		1	
	3.3×10^{-4}		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	steeper line drawn above pH 5 starting at 10%		1	AO2 AO3 3.2.4 RP3 6.3(11) 6.3(12)
	levels off at any point between 30 and 60 seconds and at 68%		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	any **three** from: • (more) accurate • it is quantitative • continuous readings • does **not** use iodine solution which could vary in volume / drop size • human error with timing is avoided	allow closer to the true value allow it is an actual value rather than an opinion **or** it is not subjective **or** colour is only qualitative allow smaller time intervals used **or** readings are taken more frequently **or** do not miss endpoint (of reaction) ignore human error unqualified	3	AO4 3.2.4 RP3 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	**Level 3:** Relevant points (reasons / causes) are identified, given in detail and logically linked to form a clear account.		5–6	AO2 3.1.5abdefg
	Level 2: Relevant points (reasons / causes) are identified, and there are attempts at logically linking. The resulting account is not fully clear.		3–4	AO2 3.1.5abdefg
	Level 1: Points are identified and stated simply, but their relevance is not clear and there is no attempt at logical linking.		1–2	AO1 3.1.5abdefg
	No relevant content		0	
	Indicative content **Water** • water in lumen is at a higher concentration than inside cell • (so) is absorbed by osmosis / diffusion • passively **or** described • down its concentration gradient • through a partially permeable membrane **Fatty acids** • fatty acids are in a higher concentration in lumen than inside cell • (so) are absorbed by diffusion • passively **or** described **Glucose** • glucose is in lower concentration in lumen than inside cell • (so) is absorbed by active transport • from an area of lower concentration to an area of higher concentration (inside the cell) **or** glucose moves up / against the concentration gradient • (which) requires energy • from respiration For **Level 3** students must identify the correct transport process for each substance.			

Total			**20**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	• a double helix	2 strands / chains that are twisted / coiled / spiralled	1	AO1 3.5.3j
	• contains four different compounds / bases		1	
		allow (long) polymer allow made up of nucleotides **or** made up of sugars, phosphates and bases		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2		all correct for **2** marks 2 correct for **1** mark	2	AO2 3.5.5cd 6.3(6) 6.3(11)
	(Herbicide resistance): 0.27(7...) or 27.7(7...)%	allow 0.28 or $\frac{5}{18}$		
	(Herbicide resistance and insect resistance): 0.55(5...) or 55.5(5...)%	allow 0.56 or $\frac{10}{18}$ or $\frac{5}{9}$		
	(Insect resistance): 0.16(6...) or 16.6(6...)%	allow 0.17 or $\frac{3}{18}$		

<table>
<tr><th>Question</th><th colspan="2">Answers</th><th>Extra information</th><th>Mark</th><th>AO / Spec. Ref.</th></tr>
<tr><td rowspan="4">04.3</td><td colspan="3">Level 2: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.</td><td>3–4</td><td>AO3
3.5.5cd</td></tr>
<tr><td colspan="3">Level 1: A conclusion, supported by some relevant reasons, is given.</td><td>1–2</td><td>AO3
3.5.5cd</td></tr>
<tr><td colspan="3">No relevant content</td><td>0</td><td></td></tr>
<tr><td colspan="3">Indicative content

For:
• GM (plant leaf area) reduced less than non-GM
• figures to support eg 49% vs 82%
• GM (plant leaf area) is 53 cm^2 more than non-GM at 5 months
• from 0–3 months GM (plant leaf area) reduced at a slower rate than non-GM or figures to support
• non-GM (plant leaf area) continued to reduce but GM increased from 3-5 months

Against:

• both (GM and non-GM plant leaf area) reduced from 0–3 months
• neither (GM and non-GM plant leaf area) increased in 5 months
• don’t know how many plants were used in the investigation
or
repeat with larger sample size
• only tested with one species of insect
or
repeat with a range of insect species
• control of variables – there may be differences in:
  • water availability
  • ion availability
  • temperature
  • light intensity
  • number of insects on each plant (some may have died or moved away or reproduced at different rates)</td><td></td><td></td></tr>
</table>

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	not all the plant cells will take up the genes from the bacteria (so) only GM cells will grow or survive **or** non-GM cells are killed		1 1	AO3 3.5.5bcd

Total			**10**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	urea	allow uric acid allow creatinine allow steroids or hormones or alcohol or named drug	1	AO1 3.4.3d

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2 (mark with 05.3)	**A**		1	AO2 3.2.3ij

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3 (mark with 05.2)	(**A**): any **two** from: • thicker wall / muscle • narrower lumen / hole • blood enters kidney (from heart)	 do **not** accept **A** / artery is thicker do **not** accept thick cell wall do **not** accept blood goes in	2	AO2 3.2.3ij

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	any **one** from: • confirm pattern in results • identify **and** ignore / repeat anomalies • to be able to calculate a mean • check / confirm for repeatability		1	AO4 3.4.3 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.5	any **two** from: • diet or fluid consumed prior to test • mass of person • sex of person • age of person • level of activity (during or before test)		2	AO4 3.4.3 6.4

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.6	increased absorption of water into blood (from the drink)	allow the blood is more dilute	1	AO3 3.4.3dfg
	is detected by pituitary gland		1	AO1
	(which causes) decrease in production of ADH **or** negative feedback on production of ADH		1	AO1 AO1
	(so) less water is reabsorbed in the kidneys **or** less water is reabsorbed into (not from) the blood **or** less water is reabsorbed from the filtrate (which increases the volume of urine)		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.7	too much water could enter cells		1	AO3 3.4.3dfg 3.1.5e
	(so) cells could swell / expand / burst		1	AO2

Total			**13**

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	(because discontinuous variation) has categories **or** specific values **or** has no gradual change between values		1	AO3 3.6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	used back legs less and less (so) they (shrunk and) were lost **or** (so) they (shrunk and) were not passed on / inherited (and) used front legs to swim or steer (in the water) (which) developed into flippers and were passed on (to offspring)	allow idea of use it or lose it for **1** mark from mp1 and mp2 allow used tail more and more (to swim) (1) (so) it became (shorter and) stronger (1)	1 1 1 1	AO2 3.6.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.3	any **one** from: • more evidence (now) for Darwin's theory **or** no evidence / proof for Lamarck's theory • more fossils discovered **or** DNA / genes better understood **or** mechanism of inheritance discovered (for Darwin's theory) • observations – people noticed that changes in their lifetime were not passed on to offspring		1	AO3 3.6.2a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.4	any **one** from: • check they are not able to interbreed successfully **or** check they cannot produce fertile offspring • analyse DNA for differences in genes		1	AO2 3.6.2c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.5	(common ancestor flew to Africa and Australia) where the environments were similar		1	AO3 3.6.2c
	example given: similar (ground level) food **or** lots of food **or** few (large) predators **or** similar ground surface to move across		1	AO3
	birds with similar advantageous features survived eg long necks to collect food at ground / high level **or** small pointed beaks to eat seeds or insects **or** flightless as less need to escape predators or to search for food **or** long powerful legs for fast running (over similar ground surfaces) to escape predators		1	AO3
	natural selection occurred in both populations with survivors passing on the genes / alleles for these (similar advantageous) features (to offspring)	allow description for natural selection	1	AO2
	(geographical isolation) led to different evolution of many characteristics so no longer able to interbreed		1	AO2

Total			12

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	temperature receptors (in skin) detect changes in temperature (receptors) send impulses to the brain / thermoregulatory centre		1 1	AO1 3.4.4ab, 3.4.2ce

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	greater loss of heat / thermal energy to (cold) environment (so) increased requirement for heat generation from respiration (which) results in increased demand for food (to be broken down) to release energy	allow increased requirement for heat generation from muscle contraction / shivering do **not** accept produced / created / make energy	1 1 1	AO3 3.2.6f 3.4.2ae 3.4.4d AO1 AO1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	(increased) sweating increased blood flow to surface of the skin	allow vasodilation allow behavioural mechanisms eg remove clothing or seek shade for **1** mark	1 1	AO2 3.4.4c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	(advantage:) any **one** from: • does not require energy from respiration (to keep warm) • less food required for respiration • greater proportion of energy from food can be used for growth or reproduction (disadvantage:) any **one** from: • greater variation in body temperature so enzymes may not work at optimum rate • not always able to find shade to cool down • (certain times of day) may be inactive and unable to (fly to) feed **or** may be inactive and unable to escape predation		1 1	AO3 3.2.4b 3.2.6bf 3.4.2ae 3.4.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	any **three** from: • (both species) the higher the (air) temperature the less time spent warming (in the sun) • species 1 spends less time warming (in the sun) compared to species 2 • species 1 shows a more rapid decline in time spent warming (in the sun after 15 °C) • at 15 °C species 1 spends 80% of time warming (in the sun) compared with species 2 which spends 100% • at 30 °C species 1 spends 14% of time warming in the sun compared with species 2 at 32%	maximum **2** marks if data not used allow tolerance of ± ½ small square allow tolerance of ± ½ small square allow other correct use of data	3	AO3 3.4.2 3.4.4

Total			**12**